

Industrial CO2 Emission Prediction Report

Report Generated: 17-09-2026 11:34:18
Record Timestamp: 17-09-2026 11:34:01
Prediction Record ID: #48

Core Telemetry Parameters

Parameter	Observed Value
Energy Consumption	16.11 kWh
Production Count	86.00 Units
Working / Shift Hour	06:00 hrs
Manufacturing Schedule	Weekend Operation

Emission Forecast & Analysis

Forecasted CO2 Emission	13.22 kg CO2
Pollution Classification	Low
Inference Methodology	Random Forest Machine Learning Model
Grid Emission Constant	0.82 kg CO2/kWh

Recommended Mitigation Measures

- Continue monitoring industrial energy consumption.
- Maintain machines regularly for energy-efficient operation.
- Continue using high-efficiency motors and variable frequency drives.
- Monitor carbon emissions periodically against baseline.

This certification report was generated by the Industrial Carbon Emission Forecasting System. Telemetry input is evaluated using Scikit-Learn Machine Learning models and Scope 2 emission standards.